
Mathematics in the Modern World

Lecture Notes

Rey Joseph Nieto

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Preface

I prepared these lecture notes for a class in *Mathematics in the Modern World* composed entirely of biology majors. The discussion, examples, and exercises are therefore written with that audience in mind. I want the notes to do more than help students complete a required mathematics course. The ideas and habits of reasoning developed here should remain useful throughout university and in later scientific, professional, and everyday work. Where practicable, the notes connect mathematics with biology, health, data, technology, and decisions that future biologists and healthcare professionals may encounter.

Each chapter is intended to accompany one three-hour class. The notes contain more examples than need to be discussed during that time, so some may be used for preparation, review, or independent study.

The notes have also been designed to comply fully with the curriculum guidelines for *Mathematics in the Modern World* issued by the Philippine Commission on Higher Education (CHED), including the prescribed learning outcomes and subject coverage [1]. Within that framework, I have tailored the explanations, applications, and supplementary examples to the needs and interests of biology students. This alignment does not imply that CHED has reviewed or endorsed these independently prepared notes.

These notes are a work in progress. Some chapters may be incomplete or absent, and existing material may be revised as the course develops. Reports of errors, corrections, and suggestions for improvement are welcome. Please send them to

rj (at) rjniето (dot) com

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Chapter 1

Mathematics in Our World

Mathematical ability is often equated with numerical fluency. The association is understandable: much of school mathematics concerns arithmetic, algebra, trigonometry, statistics, probability, and calculus. Numerical skill matters, but mathematics extends beyond calculation. It studies patterns, structures, quantities, shapes, change, relationships, and uncertainty. It also supplies a language for describing these subjects and a logical framework for reasoning about them.

Learning Outcomes

After completing this chapter, you should be able to:

- identify examples of patterns and regularities in nature and everyday life;
- explain why mathematics is more than numerical calculation;
- distinguish a mathematical model from the reality it represents;
- recognise examples of inductive and deductive reasoning;
- describe some practical, intellectual, and aesthetic dimensions of mathematics; and
- explain how mathematics contributes to biology and modern technology.

Planning an ordinary morning. Some uses of mathematics in everyday life are fairly obvious. Suppose your first class begins at 8:00 a.m. What time should you wake up to arrive on time? You might estimate how long it takes to get out of bed, take a shower, get dressed, prepare your belongings, eat breakfast, and travel to school. At first, this appears to be a simple arithmetic problem: add the estimated duration of each activity, then subtract the total from the class's starting time.

The calculation also involves more than arithmetic. You are using probability, estimation, and decision-making under uncertainty. For example, if you take public transportation, your travel time depends partly on how frequently buses, jeepneys, or tricycles pass your stop. If vehicles arrive regularly, you may expect only a short waiting time. A long wait is possible, but you may

consider it unlikely. Your estimate may also change depending on whether your class is held on a weekday or a Saturday, since traffic conditions are usually different.

You may also consider the possibility of rain, road construction, long queues, or unexpected delays. You might then add an extra fifteen or twenty minutes as a safety allowance. How large that allowance should be depends on how uncertain the journey is and how strongly you wish to avoid being late.

You have constructed a simple **mathematical model** of your morning. The model does not reproduce every detail of reality. Instead, it identifies the factors that matter most and uses them to guide a decision.



Figure 1.1: A conceptual illustration of shortest-path search across an urban road network represented as a weighted graph. The green and red markers denote the origin and destination, the white line shows the selected route, and the coloured regions indicate parts of the network explored during the search. Dijkstra's algorithm is a foundational graph-theoretic method for solving this kind of problem, although modern navigation applications employ more sophisticated variants.

Navigation applications such as Google Maps or Waze are familiar examples of systems organised around locations, routes, and travel-time estimates. At a simplified level, the mathematical toolkit for such systems includes:

- **Geometry and coordinate systems** represent locations, distances, and directions.
- **Graph theory** represents intersections as vertices and roads as weighted edges.
- **Optimisation algorithms**, including Dijkstra's algorithm [2] and A* [3], search for efficient routes through the resulting network.
- **Statistics and calculus** help estimate travel times and account for changing traffic conditions.

The estimated journey time comes from a mathematical model of location, networks, uncertainty, and change rather than from arithmetic alone.

Task

Watch the YouTube video [Why Navigation Apps Always Find the Perfect Route?](#) [4].

You will not be expected to implement Dijkstra's algorithm or A* as shown in the video. The point is to see how a familiar application uses mathematics behind the scenes. Real navigation systems extend these basic ideas to account for live traffic, road restrictions, and other practical conditions.

1.1 Patterns in Nature

Mathematics does more than help us make practical decisions. It also helps us recognise, describe, and investigate patterns—including patterns that people find beautiful.

Task

Watch the YouTube video [Nature by Numbers](#) [5].

1.2 The Golden Ratio

The video above brings together the famous Fibonacci sequence, the golden ratio, and several patterns associated with them. These ideas are closely connected, and the golden ratio provides the central thread linking much of the discussion.

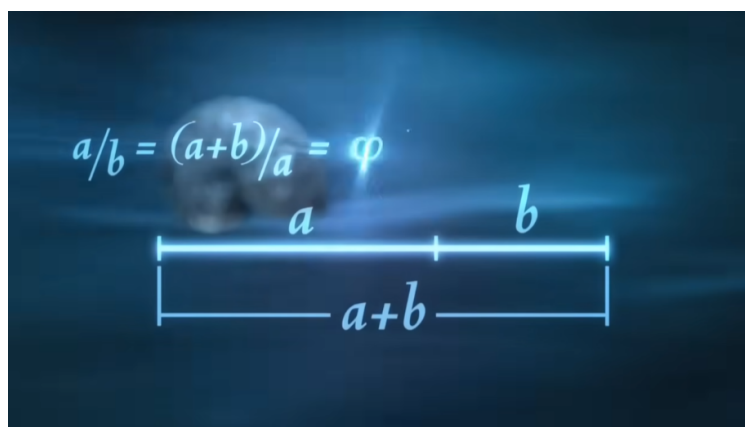
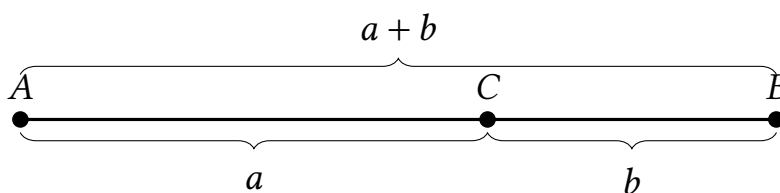


Figure 1.2: A frame from [Nature by Numbers](#) illustrating the golden ratio [5].

In Euclid's *Elements* [6], a straight line is said to be divided in the *golden ratio*¹ when

The whole is to the greater segment as the greater segment is to the lesser.

¹Euclid calls this the *extreme and mean ratio*; here we use the modern term *golden ratio*.



Let's unpack this definition. Suppose a line segment AB is divided at a point C . Let the greater segment AC have length a and the lesser segment CB have length b . The whole segment AB then has length $a + b$. Euclid's definition states that the ratios $\frac{a+b}{a}$ and $\frac{a}{b}$ are equal. Their common value is the golden ratio, usually denoted by the Greek letter φ . Thus,

$$\varphi = \frac{a+b}{a} = \frac{a}{b}.$$

This relationship allows us to calculate the numerical value of φ :

$$\varphi = \frac{a+b}{a} = 1 + \frac{b}{a} = 1 + \frac{1}{\varphi}.$$

Multiplying by φ gives $\varphi^2 - \varphi - 1 = 0$. The quadratic formula produces two roots. Because a ratio of lengths is positive, we retain the positive root:

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.6180339887.$$

1.3 The Fibonacci Sequence

Fibonacci numbers arise in mathematical descriptions of several natural patterns, particularly some arrangements of leaves, petals, and seeds. Their appearance is not universal; when these patterns do occur, they can emerge from local rules governing growth and spatial arrangement [7].



Figure 1.3: A frame from [Nature by Numbers](#) illustrating the Fibonacci sequence [5].

Spiral shells, such as those of the [nautilus](#), are often used to illustrate this idea [8]. Their shape may resemble a logarithmic spiral, but visual resemblance alone does not establish that a shell follows the Fibonacci sequence or the golden ratio exactly.

The golden ratio and the Fibonacci sequence are closely related. Before examining that relationship, let us define the sequence formally.

Definition: The Fibonacci Sequence

The *Fibonacci sequence* is the sequence $F_0, F_1, F_2, F_3, \dots$ defined by

$$F_0 = 0, \quad F_1 = 1, \quad F_n = F_{n-1} + F_{n-2} \quad \text{for } n \geq 2.$$

The first few terms are:

$$\begin{array}{ll} F_0 = 0 & F_3 = F_2 + F_1 = 1 + 1 = 2, \\ F_1 = 1 & F_4 = F_3 + F_2 = 2 + 1 = 3, \\ F_2 = F_1 + F_0 = 1 + 0 = 1 & F_5 = F_4 + F_3 = 3 + 2 = 5. \end{array}$$

Continuing this process, the sequence begins

$$0, 1, 1, 2, 3, 5, 8, 13, 21, \dots$$

Each term after the first two is the sum of the two preceding terms. Now consider ratios of consecutive nonzero terms:

$$\begin{array}{lll} \frac{F_3}{F_2} = \frac{2}{1} = 2, & \frac{F_4}{F_3} = \frac{3}{2} = 1.5, & \frac{F_5}{F_4} = \frac{5}{3} \approx 1.667, \\ \frac{F_6}{F_5} = \frac{8}{5} = 1.6, & \frac{F_7}{F_6} = \frac{13}{8} = 1.625, & \frac{F_8}{F_7} = \frac{21}{13} \approx 1.615. \end{array}$$

The ratios alternate above and below 1.618, drawing progressively closer to the golden ratio. This suggests a natural question: do the ratios actually approach φ ?

From Pattern to Proof. Computing several ratios and using the pattern to form a conjecture is an example of **inductive reasoning**. The evidence suggests a general conclusion, but finitely many examples cannot prove a claim about the entire infinite sequence. **Deductive reasoning** instead begins with definitions, accepted facts, or clearly stated assumptions and uses valid logical steps to reach a conclusion. The distinction matters: a pattern may suggest what is true, while a proof explains why it must be true.

Fibonacci and the Golden Ratio (Non-examinable)

Recall that

$$F_n = F_{n-1} + F_{n-2}.$$

We are interested in the behaviour of F_n/F_{n-1} as n becomes large. Assume that this ratio approaches a positive limit L :

$$L = \lim_{n \rightarrow \infty} \frac{F_n}{F_{n-1}}.$$

Dividing the recurrence relation by F_{n-1} gives

$$\frac{F_n}{F_{n-1}} = 1 + \frac{F_{n-2}}{F_{n-1}} = 1 + \frac{1}{F_{n-1}/F_{n-2}}.$$

The shifted ratio F_{n-1}/F_{n-2} has the same limit L . Taking limits therefore gives

$$L = 1 + \frac{1}{L}.$$

Because $L > 0$, solving $L^2 - L - 1 = 0$ yields

$$L = \frac{1 + \sqrt{5}}{2} = \varphi.$$

A complete proof must also establish that the limit exists; that additional step is beyond the scope of this course.²

1.4 The Golden Angle and a Word of Caution

In botany, **phyllotaxis** describes how leaves, seeds, or other plant organs are arranged. In some spiral arrangements, successive organs form an angle close to the **golden angle**

$$\frac{360^\circ}{\varphi^2} \approx 137.5^\circ.$$

This spacing is associated with the familiar Fibonacci-like spirals seen in some flower heads and other plant structures.

This pattern is a genuine subject of research in mathematical biology [7], but it should not be treated as a universal rule. Not every plant follows a golden-angle pattern, and plants do not

²(Non-examinable) The argument above is only one route to the result. Other methods, including techniques from linear algebra, lead to the same conclusion. A correct and clearly explained method can receive full credit even if it differs from the one presented in class. You are not expected to discover a more advanced approach, but you should feel free to use one if you can justify it.



Figure 1.4: A frame from [Nature by Numbers](#) illustrating a golden-angle pattern in a sunflower head [5].

calculate φ . Rather, the pattern can emerge from biological growth processes and interactions among developing structures in limited space.

A word of caution. The golden ratio appears in some natural patterns, but claims about it are often exaggerated. Faces, paintings, buildings, shells, and galaxies are frequently described as “golden” without sufficient evidence.



Figure 1.5: The Parthenon in Greece [9]. Although the building is often said to have been designed around the golden ratio, historical and geometrical evidence does not support that claim [10–12].

A shape that looks approximately golden is not evidence that the ratio was deliberately used or that it caused the shape to form. A mathematical claim should be supported by clear definitions, reproducible measurements, and—when biology is involved—a plausible biological mechanism.

1.5 Mathematics and Biomedical Imaging

The golden ratio is an attractive example, but its practical importance is easily overstated. Biomedical imaging offers a more direct example of mathematics at work.



Figure 1.6: An illustration of normal vision (left) and vision affected by advanced diabetic retinopathy (right). Early DR may cause no noticeable symptoms [13].

Figure 1.6 illustrates what a person might see with a healthy eye and with an eye affected by **diabetic retinopathy** (DR), a complication of diabetes. Diabetes can damage the small blood vessels in the retina, the light-sensitive layer at the back of the eye. In advanced cases, these vessels may leak or bleed, causing blurred vision, dark floating spots, or even blindness [13]. To screen for the disease, clinicians examine the retina using **fundus photographs**—colour photographs of the back of the eye—such as those shown in Figure 1.7 [14].

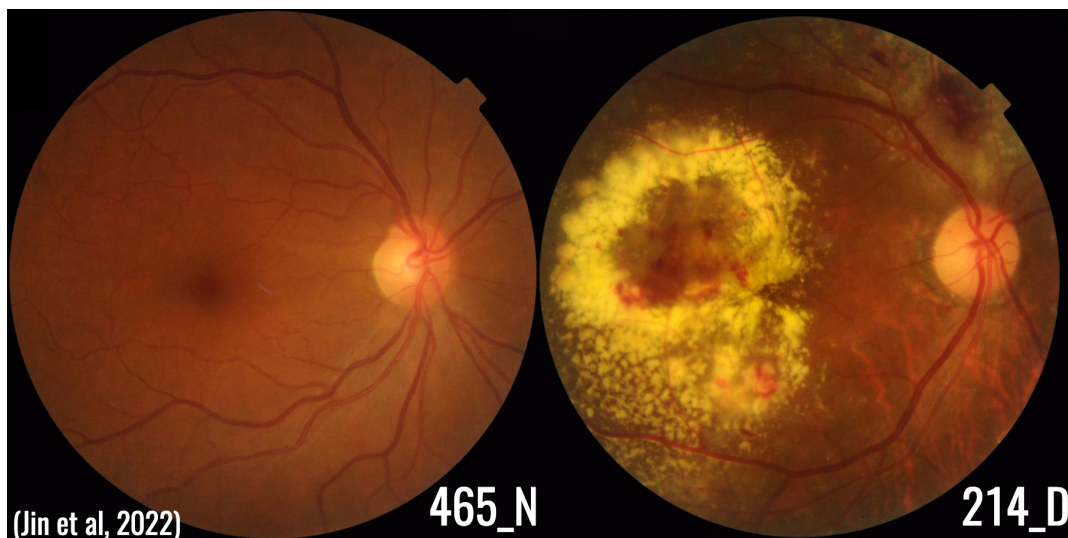


Figure 1.7: Fundus photographs from the FIVES dataset: a normal retina (left) and a retina affected by diabetic retinopathy (right) [15].

In this example, the image on the right appears much more irregular than the one on the left. It contains large yellow regions and other visible abnormalities. This particular case is relatively

easy to distinguish by inspection, but not every case is so obvious. The signs of early disease can be subtle, and a diagnosis must always be made by a qualified clinician.

Some biology graduates go on to work in medicine and other health professions. In settings where many patients require screening, computers can help organise images and flag cases that may need closer attention. Such systems do not replace clinicians. Their purpose is to help clinicians review large collections of images and identify patients who may require further assessment.

The diseased image in Figure 1.7 might be described informally as looking “messier”. A computer cannot use such a description directly; it needs features that can be measured numerically.

One approach is **topological data analysis** (TDA), a mathematical framework that can be combined with machine learning and computer vision [16]. In my bachelor’s thesis [17], I drew on ideas from several mathematical fields, including linear algebra, graph theory, algebraic topology, and category theory, to convert fundus images into numerical signatures. These signatures can then be supplied to a machine-learning system that classifies an image as either normal or affected by diabetic retinopathy. Figure 1.8 summarises this process.

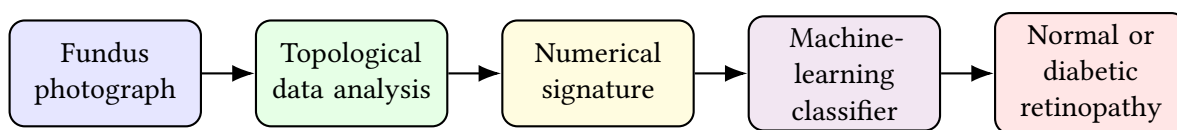


Figure 1.8: A simplified overview of the TDA-assisted retinal-image classification workflow [17].

The underlying idea is manageable. TDA identifies structural features in an image, such as separate components and loops, and summarises them as numerical information [16]. In the retinal workflow used here, that information forms a **topological signature** [17].

Once each image has a numerical signature, the signatures of different groups can be compared statistically. They can also be supplied to a machine-learning model, such as an artificial neural network.

This does not mean that TDA can diagnose diabetic retinopathy on its own. Digital images are already composed of numerical pixel values, and many computational methods analyse those values directly. TDA provides an additional description that focuses on shape and connectedness. Researchers must therefore test whether this additional information makes a computer-assisted screening system more accurate, robust, or reliable.

Machine Learning and Modern Artificial Intelligence. Machine learning provides much of the foundation for modern artificial intelligence, and many machine-learning systems employ **artificial neural networks**. Despite their biologically inspired name, these are mathematical and computational models rather than literal replicas of a brain.

Building and training these models draws on several disciplines. Computer science supplies the algorithms and software; linear algebra represents data and the connections within a network; calculus helps adjust those connections during training; and probability and statistics help describe uncertainty, detect patterns, and evaluate performance [18]. Their operation depends on computation together with several branches of mathematics.

Key Takeaway

The retinal example uses many of the ideas introduced at the start of the chapter. Shapes and structures in an image are represented by numbers, compared across groups, and examined for relationships with disease. Statistics and machine learning help account for uncertainty when an image is classified.

This is an example of mathematics turning an abstract description of shape into a practical tool that can support medical screening and patient care.

1.6 Is Mathematics an Exact Science?

Mathematics is often described as exact, but that statement needs an important qualification. **Pure mathematics can be exact within its definitions and assumptions.** Once we agree on the rules, a valid proof establishes its conclusion with logical certainty. The real world, however, does not arrive with perfectly defined objects, flawless measurements, or assumptions that are always true.

Reporting a Person's Height

Suppose someone reports a height of 5 feet 6 inches. This normally does not mean that the person's height is *exactly* 66 inches. The measurement may have been rounded to the nearest inch; it may also vary slightly with posture, footwear, and the measuring instrument. Nevertheless, "5 feet 6 inches" is usually precise enough for an ordinary conversation or a form asking for someone's height. Greater precision would add detail, but it would not necessarily be useful for the purpose at hand.

The same principle applies when mathematics is used to study traffic, populations, disease, climate, finance, or biological systems. A **mathematical model** keeps the features judged relevant to a question and leaves others out. It is therefore a simplified representation of reality, not reality itself. Its conclusions depend both on sound mathematics and on how well its assumptions and data represent the situation being studied.

This does not make applied mathematics unreliable. It means that a model should be judged in context: What is it intended to do? How accurate must it be for that purpose? What assumptions does it make? How uncertain are its measurements and predictions? A metro map is not

geographically exact, for example, but it can still be an excellent model for finding the correct train.

Exact Reasoning. Mathematics allows us to reason precisely about a chosen model. It does not guarantee that the model is a perfect copy of the world. A useful model is often not the one containing every possible detail, but the simplest one that is sufficiently accurate for its intended purpose.

1.7 Mathematics and Collaboration

Mathematicians are trained to work with quantities, patterns, structures, and carefully stated assumptions. That training does not supply all the biological knowledge needed to frame a problem well. As the familiar saying warns, “If all you have is a hammer, everything looks like a nail.” A mathematician who approaches a biological problem using mathematics alone may construct an elegant model while overlooking something biologically important.

Effective applications therefore usually require collaboration with **domain experts**—people with specialised knowledge of the subject being investigated. In biology, these experts may include laboratory researchers, field biologists, clinicians, epidemiologists, and public-health specialists. They help determine which questions matter, which assumptions are reasonable, what data can be collected, and whether a mathematical conclusion makes biological sense.

Mathematical competence does not confer expertise in biology, medicine, or any other field to which mathematics is applied. A calculation can be correct even when the model behind it is biologically inappropriate. Biologists, in turn, benefit from enough mathematics to question a model’s assumptions, interpret its results, and recognise its limitations. Good interdisciplinary work requires knowing when to ask questions, collaborate, or defer to someone with the relevant expertise.³

Mathematicians and statisticians can translate these questions into models, analyse the resulting data, and quantify uncertainty. Computer scientists can develop the algorithms and systems needed to process large or complex datasets. Together, domain expertise, mathematics, and computer science form a common foundation for **data science**, as illustrated in Figure 1.9. Data science draws on all three areas to extract meaningful information from data and support better decisions [20].

A biologist need not become a mathematician to benefit from a working understanding of mathematical ideas. That understanding makes it easier to communicate with quantitative

³(Non-examinable) Philosophers use the term **epistemic trespassing** for passing judgement on matters outside one’s field of expertise. Awareness of this risk encourages intellectual humility: expertise in one discipline should not be assumed to transfer automatically to another [19].

specialists and to recognise when a biological question can be measured, modelled, or investigated in a new way.⁴

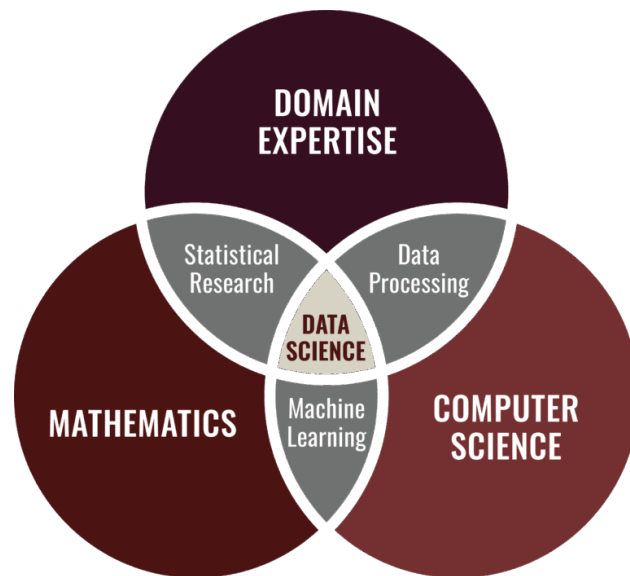


Figure 1.9: A common view of data science as the intersection of domain expertise, mathematics, and computer science. The overlapping areas also support statistical research, data processing, and machine learning [20].

Some biologists specialise in mathematical or computational work, whereas others concentrate on laboratory, clinical, or field research. Progress often depends on people with these different forms of expertise understanding one another well enough to work on the same problem. Mathematics provides one of the shared languages that makes such work possible.

1.8 What It Means to Think Mathematically

Being mathematically capable requires more than performing calculations. It includes the ability to recognise patterns and relationships, identify relevant information, construct and evaluate models, reason logically from assumptions, account for uncertainty, compare possible solutions, and explain why a conclusion should be accepted.

Numbers remain indispensable, but they are only part of the story. Mathematics includes numerical techniques, but it also organises information, examines relationships, solves problems, and supports careful reasoning about the world.

⁴My academic background is primarily in mathematics and computer science. To address the biological aspects of my thesis, I relied on well-established findings from biologists. This experience illustrates why many of today's important discoveries are collaborative efforts involving domain experts, mathematicians, and computer scientists. Although developing expertise in one field remains important, interdisciplinary work becomes much easier when each participant also has some familiarity with the knowledge, methods, and language of the other disciplines.

1.9 Exercises

These questions require you to apply the ideas developed in the chapter.

1. **Model an ordinary journey.** List the factors that could affect how long it takes you to travel to class. Which factors would you include in a useful model, and which details could you reasonably leave out?
2. **Investigate a pattern.** Calculate the next three Fibonacci numbers and the ratios of successive terms. Do the new ratios support the conjecture that they approach ϕ ? Explain why this is evidence rather than a complete proof.
3. **Describe an image mathematically.** Compare the two retinal photographs above. Suggest one visible feature that could be measured numerically. Explain what information your proposed measurement captures and what it might overlook.
4. **Question a claim.** Find an everyday or biological claim involving a mathematical pattern. What definition, measurements, or evidence would you require before accepting it?
5. **Defend a broader view of mathematics.** A classmate says that mathematics is merely a collection of formulae for obtaining numerical answers. Write a short response challenging this view. Use two examples from the chapter to explain how mathematics can be expressed, represented, and used, and identify at least one practical, intellectual, or aesthetic dimension of the subject.
6. **Apply mathematics responsibly.** Suppose the developers of a computer-assisted retinal-screening system know that its performance deteriorates when photographs are blurry, but advertise only its overall accuracy. What information should they disclose? Explain how honesty, integrity, and awareness of a model's limitations should affect the use of such a system in healthcare.

Closing Reflection

What decisions have you already made today that involved mathematics, even though you did not consciously perform a written calculation?

Chapter 2

The Language of Mathematics

Unfamiliar notation can make mathematics appear more intimidating than it really is. Mathematical symbols are not meant to conceal ideas. Rather, they allow us to express those ideas efficiently and with less ambiguity than ordinary language sometimes permits.

Learning Outcomes

After completing this chapter, you should be able to:

- distinguish among mathematical expressions, open sentences, and closed sentences, and translate elementary statements between words and symbols;
- read common mathematical symbols and apply standard conventions, including the order of operations;
- use sets, ordered pairs, relations, functions, and binary operations in elementary examples;
- interpret elementary logical connectives and quantifiers, and distinguish a conditional statement from its converse and negation;
- explain how formality reduces ambiguity in mathematical communication; and
- recognise these ideas in biology and everyday life.

Like any language, mathematics has a vocabulary, a grammar, and a set of conventions. Its vocabulary consists of symbols and specialised terms. Its grammar determines which arrangements of symbols are meaningful. Its conventions allow different readers to interpret the same notation consistently.

Task

Watch the YouTube video [The Language of Maths](#) [21].¹

Mathematical language is valued for its **precision**: it allows us to make distinctions that ordinary language may leave ambiguous. It is also **concise**, since it can communicate an idea

¹(Non-examinable) In American English, *mathematics* is usually shortened to *math*. In British and Australian English, it is usually shortened to *maths*; hence the title of the video.

using relatively few symbols. Finally, it is **powerful**: those symbols can describe complicated structures and relationships in a form that can be analysed logically.

These advantages do not mean that mathematical writing should consist only of symbols. Good mathematical communication combines notation with clear prose. Symbols compress ideas; the surrounding words explain what those ideas mean and why they matter.

Translating a Temperature Condition

Consider the sentence “The body temperature T is at least 38°C .” The phrase **at least** includes 38, so we write

$$T \geq 38.$$

The symbolic form is **concise**: it expresses the condition using only a few symbols. It is also **precise** once we have established that T denotes temperature measured in degrees Celsius and that $\geq$ means “is greater than or equal to.” In contrast, “the body temperature is greater than 38°C ” is written as

$$T > 38.$$

Removing the small bar beneath $>$ excludes a temperature of exactly 38°C .

It is **powerful** because the condition can now be used in calculations, compared with other inequalities, and manipulated using the rules of algebra.

2.1 Expressions and Sentences

The distinction between a mathematical **expression** and a mathematical **sentence** is similar to the distinction between a noun phrase and a complete sentence in English.

Definitions: Expressions and Sentences

A **mathematical expression** is a meaningful arrangement of symbols that represents a mathematical object, value, or quantity. An expression does not, by itself, make a claim that can be judged true or false.

A **mathematical sentence** is a meaningful arrangement of symbols that makes a mathematical claim. A **closed sentence**, also called a **logical proposition**, can be judged either true or false.

An assertion whose truth depends on the value assigned to an unspecified variable is called an **open sentence**. Assigning a value to the variable, or introducing a phrase such as “for every” or “there exists,” can turn an open sentence into a closed sentence.

A useful first test is to read the notation aloud. A mathematical sentence makes a complete claim, whereas an expression does not. A sentence is open when one or more of its variables still

need values before the claim can be judged true or false. It is closed when no such assignment is needed. Note that the appearance of a variable does not automatically make a sentence open. In

$$\forall x \in \mathbb{R}, \quad x^2 \geq 0,$$

the phrase “for every x ” tells us how x is to be interpreted, so the statement can be judged true or false as written.²

Notation	Classification	Reason
$b^2 - 4ac$	expression	represents the discriminant of $ax^2 + bx + c$ but makes no assertion
$\frac{y_2 - y_1}{x_2 - x_1}$	expression	may represent the slope of a line but makes no assertion
$x + 1 = 2$	open sentence	its truth depends on the value assigned to x
$2x + 3 > 7$	open sentence	its truth depends on the value assigned to x
$1 + 1 = 2$	true closed sentence	it makes a true assertion and contains no unspecified variables
$1 + 1 = 3$	false closed sentence	it makes a false assertion and contains no unspecified variables
$\forall x \in \mathbb{R}, x^2 \geq 0$	true closed sentence	the variable is governed by “for every”, and every real number has a nonnegative square
$\exists x \in \mathbb{R}, x^2 = -1$	false closed sentence	no real number has a negative square

Mathematical sentences often join expressions using relation symbols such as $=$, $\neq$, $<$, $\leq$, and $\in$. In the sentence $1 + 1 = 2$, for example, the equality sign asserts that the expressions on its two sides represent the same number.

2.1.1 Reading Common Mathematical Symbols

Symbols should be read as meaningful words or phrases, not as decorative marks. The following table lists some symbols that will appear throughout these notes.

²(Non-examinable) Expressions, sentences, variables, and quantifiers are studied formally in **mathematical logic**. Much of the notation used here belongs to **first-order predicate logic**, also called **first-order predicate calculus**. A **Hilbert-style calculus** is one formal system for constructing proofs in logic [22]. Here, *calculus* means a systematic method of calculation or reasoning, not necessarily differentiation and integration. Differential and integral calculus became so influential that they nearly monopolised the word, but many other calculi exist. The plural of *calculus* is *calculi*.

Symbol	Meaning	Read as	Example
$=$	equality	equals	$2 + 3 = 5$
$\neq$	non-equality	is not equal to	$2 + 3 \neq 6$
$<$	strict inequality	is less than	$3 < 5$
$>$	strict inequality	is greater than	$8 > 4$
$\leq$	inequality	is less than or equal to	$x \leq 10$
$\geq$	inequality	is greater than or equal to	$x \geq 0$
$\approx$	approximate equality	is approximately equal to	$\pi \approx 3.14$
$\pm$	two possible signs	plus or minus	$x = \pm 2$
$\in$	set membership	is an element of	$3 \in \mathbb{N}$
$\notin$	set non-membership	is not an element of	$-1 \notin \mathbb{N}$
$\mathbb{N}$	natural numbers	the natural numbers	$5 \in \mathbb{N}$
$\mathbb{Z}$	integers	the integers	$-3 \in \mathbb{Z}$
$\mathbb{Q}$	rational numbers	the rational numbers	$\frac{3}{4} \in \mathbb{Q}$
$\mathbb{R}$	real numbers	the real numbers	$\sqrt{2} \in \mathbb{R}$
$\mathbb{C}$	complex numbers	the complex numbers	$2 + 3i \in \mathbb{C}$
$\subseteq$	subset	is a subset of	$\mathbb{N} \subseteq \mathbb{Z}$
$\cup$	union of sets	union	$A \cup B$
$\cap$	intersection of sets	intersection	$A \cap B$
$\forall$	universal quantifier	for every	$\forall x \in \mathbb{R}, x^2 \geq 0$
$\exists$	existential quantifier	there exists	$\exists x \in \mathbb{R}, x^2 = 4$
$\Rightarrow$	implication	implies	$x > 2 \Rightarrow x > 0$
$\Leftrightarrow$	equivalence	if and only if	$x = 0 \Leftrightarrow x^2 = 0$
$\therefore$	conclusion	therefore	$x = 2, \therefore x^2 = 4$
$:=$	definition	is defined as	$f(x) := x^2 + 1$
$\sum$	summation	the sum of	$\sum_{i=1}^n x_i$
$\prod$	repeated multiplication	the product of	$\prod_{i=1}^n x_i$
$\propto$	proportionality	is proportional to	$y \propto x$

Do I Need to Memorise All These Symbols? No. You may write a mathematical statement in words, provided that its meaning is clear and correct. Learning the common symbols is nevertheless worthwhile: they make mathematical writing shorter, save time when you write solutions, and help you read textbooks and scientific material more easily. Familiarity will develop through use, so you do not need to master every symbol at once. This compactness is also one reason why some mathematics textbooks are physically thinner than textbooks in subjects such as biology: a short line of notation can express what might otherwise require several sentences. If every symbol and intermediate step were repeatedly written out in full, the same mathematical material could occupy many more pages.

2.1.2 Translating Between Words and Symbols

Translating between ordinary language and mathematical notation requires attention to meaning rather than word-for-symbol substitution.

Additional Translation Examples.

Words	Symbols
x is positive	$x > 0$
n is no more than 20	$n \leq 20$
p is between 0 and 1, inclusive	$0 \leq p \leq 1$
Population density d is population N divided by area A	$d = \frac{N}{A}$
x belongs to the set A	$x \in A$
A and B have no common elements	$A \cap B = \emptyset$
There is a real number whose square is 4	$\exists x \in \mathbb{R} : x^2 = 4$

For example, the order of words matters when translating the phrase “less than”. The phrase “five less than x ” means “ $x - 5$ ”, and not “ $5 - x$ ”.

2.2 Conventions in Mathematical Language

Mathematical conventions are agreed ways of writing and interpreting mathematics. Without them, two readers could assign different meanings to the same line of notation.

2.2.1 Order of Operations

One familiar convention is the order of operations, often remembered using **PEMDAS**: parentheses, exponents, multiplication, division, addition, and subtraction.

Multiplication and division have equal priority, as do addition and subtraction. Operations of equal priority are performed from left to right.

Example: Applying the Order of Operations. Consider

$$\begin{aligned}
 1 + 2 - 3(4 + 5) &= 1 + 2 - 3(9) && \text{parentheses} \\
 &= 1 + 2 - 27 && \text{multiplication} \\
 &= 3 - 27 && \text{addition} \\
 &= -24. && \text{subtraction}
 \end{aligned}$$

Following the agreed convention gives a unique result. Performing the subtraction before the multiplication would produce a different answer.

2.2.2 Other Common Conventions

- If n is a positive integer, then

$$x^n = \underbrace{x \cdot x \cdots x}_{n \text{ factors}}.$$

- Symbols written next to one another usually indicate multiplication: $3x = 3 \cdot x$ and $ab = a \cdot b$.
- A fraction bar groups the entire numerator and denominator:

$$\frac{a+b}{c+d} = (a+b) \div (c+d).$$

- Parentheses can change the quantity to which an operation applies:

$$-3^2 = -(3^2) = -9, \quad (-3)^2 = 9.$$

- The notation $f(x)$ usually denotes the value of the function f at x . It does not ordinarily mean f multiplied by x .
- The equality sign expresses sameness of value. Every expression in a chain such as $a = b = c$ must represent the same object or value.
- The approximation sign $\approx$ should be used when two quantities are close but not exactly equal. For example, $\pi \approx 3.14$, not $\pi = 3.14$.

Conventions are not laws of nature. They are shared agreements that make mathematical communication efficient and consistent. With these conventions in place, we can turn to several structures that recur throughout mathematics.

2.3 Sets, Relations, Functions, and Binary Operations

Sets collect objects; relations describe connections among them; functions assign outputs to inputs; and binary operations combine two objects to produce a third. We begin with sets.

2.3.1 Sets

For this course, the following informal definition is sufficient.³

³(Non-examinable) This is the usual informal definition from **naïve set theory**. In more advanced mathematics, allowing any verbal description to define a set can produce contradictions, including Russell's paradox [23]. Modern axiomatic set theory restricts how sets may be constructed in order to avoid such contradictions.

Definition: Set

For the purposes of this course, a **set** is a well-defined collection of distinct objects. The objects belonging to the set are called its **elements**.

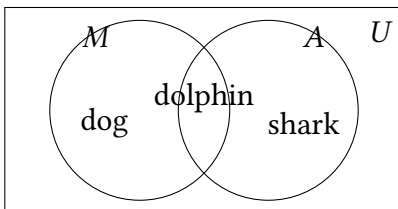
Set Notation. Sets are usually denoted by capital letters. For example, let $D = \{A, C, G, T\}$, where A, C, G , and T denote adenine, cytosine, guanine, and thymine, respectively. Then $A \in D$, meaning that adenine is an element of D .

Repeated elements are listed only once, and their order does not matter. Thus, $\{1, 2, 2, 3\} = \{3, 2, 1\} = \{1, 2, 3\}$.

If every element of a set A is also an element of a set B , we write $A \subseteq B$ and say that A is a **subset** of B . The set containing no elements is called the **empty set** and is denoted by $\emptyset$.

Set Operations. Two important operations on sets are the **union** $A \cup B$, which contains the elements that belong to A , to B , or to both. The **intersection** $A \cap B$ contains only the elements common to both sets.

Example: A Venn Diagram. A Venn diagram represents sets as regions. Let the universal set U contain a dog, a dolphin, and a shark. Let M contain the mammals in U , and let A contain the aquatic animals. The dolphin belongs to both sets, so it is placed in their overlap.



Thus, $M \cap A = \{\text{dolphin}\}$ and $M \cup A = \{\text{dog, dolphin, shark}\}$.

2.3.2 Ordered Pairs and Cartesian Products

Example: Ordered Pairs and the Cartesian Plane. An **ordered pair** (a, b) records both its entries and their order; in general, $(a, b) \neq (b, a)$.

The **Cartesian product** of two sets A and B is

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}.$$

It is the set of all ordered pairs whose first entry belongs to A and whose second entry belongs to B .

The familiar Cartesian plane is $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$. Each point (x, y) represents one location: x gives its horizontal position and y gives its vertical position.

2.3.3 Relations

Definition: Relation

A **relation from A to B** is a subset of $A \times B$. It records which elements of A are related to which elements of B .

Relations may also be defined on a single set. In an ecological network, for example, “preys upon” is a relation on a set of species.

Example: A Relation as a Dietary Model. Let A contain a dog, a wolf, and a hyena, and let $B = \{\text{carnivore, herbivore}\}$. In a deliberately simplified model, consider the relation

$$R = \{(\text{dog, carnivore}), (\text{dog, herbivore}), (\text{wolf, carnivore}), (\text{hyena, carnivore})\}.$$

Here, the dog is paired with two elements of B .

Now let $B' = \{\text{carnivore, herbivore, omnivore}\}$ and define

$$R' = \{(\text{dog, omnivore}), (\text{wolf, carnivore}), (\text{hyena, carnivore})\}.$$

Unlike R , the relation R' pairs every element of A with exactly one element of B' . Relations with this property form an important special class called **functions**, which we define in the next section.

The categories deliberately simplify real animal diets. They are useful here because they make the difference between R and R' easy to see.

Example: Family Relationships as a Relation. Let P be a set of people. We can define a relation R on P by declaring that aRb if and only if a is a relative of b . Thus, $(a, b) \in R$ means that person a is a relative of person b .

Example: Number Systems as a Relation. Let $A = \mathbb{C}$ and $B = \{\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}\}$. Define xRS to mean that $x \in S$. For example, $2R\mathbb{N}$, $2R\mathbb{Z}$, and $2R\mathbb{R}$, while $-3R\mathbb{Z}$ and $\sqrt{2}R\mathbb{R}$.

2.3.4 Functions

Functions are a special type of relation.

Definition: Function

A **function f** from a set A to a set B , written $f: A \rightarrow B$, is a relation that assigns **every** element of A to **exactly one** element of B .

The set A is called the **domain**, and B is called the **codomain**. The set of outputs actually produced by the function is its **range**. The range is a subset of the codomain.

The phrase “exactly one” is essential. Different inputs may have the same output, but a single input cannot be assigned two different outputs by the same function. A function $f: A \rightarrow B$ must also assign an output to every element of its stated domain A .

Example: DNA Base Pairing as a Function. Let $D = \{A, C, G, T\}$, where the letters denote adenine, cytosine, guanine, and thymine, respectively. Figure 2.1 shows these bases within the familiar double-helix structure.



Figure 2.1: An illustration of the DNA double helix. Its complementary base-pairing rules can be modelled as a function [24].

Complementary base pairing assigns adenine to thymine and cytosine to guanine. This rule defines a function $f: D \rightarrow D$ by

$$f(A) = T, \quad f(T) = A, \quad f(C) = G, \quad f(G) = C.$$

Every input base has exactly one complementary base, so this rule is a function.

2.3.5 Binary Operations

Definition: Binary Operation

A **binary operation** on a set A is a function $\star: A \times A \rightarrow A$. It takes two elements of A , combines them according to a specified rule, and produces another element of A .

Examples of Binary Operations. Addition. Addition is a binary operation on the integers, $+: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$, because the sum of any two integers is another integer.

Set union. Similarly, if $\mathcal{P}(U)$ is the set of all subsets of a fixed set U , then union is a binary operation $\cup: \mathcal{P}(U) \times \mathcal{P}(U) \rightarrow \mathcal{P}(U)$.

Example: When an Operation Is Not Binary. The underlying set matters. Subtraction is not a binary operation on the positive integers because $2 - 3 = -1$, which is not positive. Division is not a binary operation on $\mathbb{R}$ because division by zero is undefined.

Example: A Custom Binary Operation. The meaning of an operation symbol comes from its definition. An unfamiliar symbol need not represent one of the familiar arithmetic operations.

For example, define an operation $\star$ on the real numbers by

$$a \star b = a + 2b.$$

Applying the stated rule gives

$$3 \star 4 = 3 + 2(4) = 11.$$

2.4 Elementary Logic

Sets, relations, functions, and operations describe mathematical objects and how they are connected. Logic describes the claims we make about those objects.

Logic studies valid reasoning: how statements are combined and which conclusions follow from them.

Scope of this section. We need enough logic to read the basic connectives and quantifiers, interpret simple statements, complete a two-proposition truth table, and distinguish a conditional from its converse. Longer truth tables and formal proofs of logical equivalence lie beyond our scope.

Definition: Logical Proposition

A **logical proposition** is a closed sentence that is either true or false. Under a fixed interpretation, it cannot be both true and false.⁴

For example, “A dog is a mammal” is a logical proposition. The expression $x + 1$ makes no claim, while the open sentence $x + 1 = 2$ cannot be judged true or false until x is specified or quantified.

⁴(Non-examinable) In this section, **logical proposition** means a statement that can be judged true or false. Mathematical writers also use **Proposition** as a label for a proved result. A theorem is usually presented as a major result; a proposition as a useful stand-alone result that the author considers less central; and a lemma as a result used to prove a later theorem or proposition. These labels are not strict rankings and may vary from one text to another. Theorems, propositions, and lemmas are all logical propositions because they are statements that can be judged true or false.

2.4.1 Logical Connectives

Simple logical propositions can be joined to form compound logical propositions. Let P and Q denote logical propositions.

Symbol	Name	Reading	When it is true
$\neg P$	negation	not P	when P is false
$P \wedge Q$	conjunction	P and Q	when both P and Q are true
$P \vee Q$	disjunction	P or Q	when at least one of P and Q is true
$P \Rightarrow Q$	conditional	if P , then Q	except when P is true and Q is false
$P \Leftrightarrow Q$	biconditional	P if and only if Q	when P and Q have the same truth value

In mathematics, the word **or** is normally inclusive. Thus, $P \vee Q$ is true when P is true, when Q is true, or when both are true.

Example: A Truth Table for the Logical Connectives. A **truth table** displays the truth value of a compound logical proposition for every possible combination of truth values of its components.

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Example: A Conditional and Its Converse. Let P mean “the organism is a dog” and Q mean “the organism is a mammal.” The conditional $P \Rightarrow Q$ says, “If the organism is a dog, then it is a mammal.”

Under the usual biological classification, this statement is true. Its **converse**, $Q \Rightarrow P$, says, “If the organism is a mammal, then it is a dog.”

The converse is false because many mammals are not dogs. Thus, a conditional statement and its converse need not have the same truth value.

Implication Is Not Necessarily Causation. The notation $P \Rightarrow Q$ means that whenever P is true, Q must also be true. It does not, by itself, explain why.

For example, let P mean “I bang my head against a wall” and let Q mean “my head hurts.” The implication $P \Rightarrow Q$ describes a causal relationship: the impact causes the pain. Now let R mean “I have a bump on my forehead.” After the same accident, $R \Rightarrow Q$ may also be true in a particular account of the accident, but it does not follow that the bump caused the pain. The impact may have

caused both. An implication can occur alongside causation, but it does not establish causation by itself. Nor does $P \Rightarrow Q$ imply that the converse $Q \Rightarrow P$ is true.

2.4.2 Quantifiers

Quantifiers state whether a condition holds for every element, for at least one element, or for exactly one element.

Symbol	Reading
$\forall$	for every, or for all
$\exists$	there exists at least one
$\exists!$	there exists exactly one

Example: Quantifying DNA Base Pairing. The DNA complementary-base function satisfies

$$\forall x \in D, \quad f(f(x)) = x.$$

This says that, for every DNA base x in D , taking the complement twice returns the original base.

Example: Why the Domain Matters. The domain of a quantifier matters. Consider the logical proposition

$$\forall x \in \mathbb{Z}, \quad \exists y \in \mathbb{Z} \text{ such that } x + y = 0.$$

This proposition is true. Given any integer x , we may choose $y = -x$, which is also an integer. The number y is called the **additive inverse** of x .

By contrast, the proposition

$$\forall x \in \mathbb{N}, \quad \exists y \in \mathbb{N} \text{ such that } x + y = 0$$

is false when $\mathbb{N}$ denotes the natural or counting numbers. For example, if $x = 1$, the only real number satisfying $1 + y = 0$ is $y = -1$, and $-1 \notin \mathbb{N}$.

The same equation can therefore lead to different truth values when the permitted domain changes.

Example: Biological Classification Depends on the Domain. The same issue arises in biological classification. Consider the traditional five-kingdom model, and let $\mathcal{L}$ be the set of living cellular organisms covered by that model [25]. Let

$$\mathcal{K} = \{\text{Monera, Protista, Fungi, Plantae, Animalia}\},$$

where each kingdom is itself treated as a set of organisms. The claim that every organism receives a kingdom classification can be expressed as

$$\forall x \in \mathcal{L}, \quad \exists Y \in \mathcal{K} \text{ such that } x \in Y.$$

Now suppose we broaden the domain to a set $\mathcal{G}$ containing all biological entities that possess a genome, while leaving $\mathcal{K}$ unchanged. The corresponding proposition

$$\forall x \in \mathcal{G}, \quad \exists Y \in \mathcal{K} \text{ such that } x \in Y$$

is false because the enlarged domain includes viruses, which are not assigned to any of the five kingdoms.

The varicella-zoster virus—which causes chickenpox and shingles—provides one counterexample. It has a double-stranded DNA genome [26], but it does not belong to Monera, Protista, Fungi, Plantae, or Animalia.

Defining the domain as “entities that possess a genome” also allows us to avoid the separate and contested question of whether viruses should be regarded as living.

Changing the domain exposes a limitation of the five-kingdom model. The quantified proposition is true for the organisms the classification was designed to cover but false after viruses are included.

Example: Negating Quantified Propositions. Negating a quantified proposition changes its quantifier:

$$\neg(\forall x, P(x)) \iff \exists x \text{ such that } \neg P(x), \quad \neg(\exists x, P(x)) \iff \forall x, \neg P(x).$$

In words, the negation of “Every organism has property P ” is not “No organism has property P .” The correct negation is the weaker statement “At least one organism does not have property P .”

Logical notation makes the structure of an argument explicit. The same precision allows computers to apply logical rules mechanically.

Task

Watch [An Introduction to Propositional Logic](#) [27].

2.5 Formality in Mathematical Language

In ordinary conversation, context, tone, and shared experience often help us interpret incomplete or ambiguous statements. Mathematics aims to reduce this dependence on context.

Formality means expressing an idea with enough structure that its objects, assumptions, and logical relationships can be interpreted consistently.

Example: Formalising a Classification Rule. Consider the informal sentence “Each animal has one classification.” This statement does not identify the animals or the possible classifications. Moreover, the word “one” might mean “at least one” or “exactly one”.

In the dietary model, the sets A and B' and the relation R' allow us to write the intended claim as

$$\forall a \in A, \quad \exists! b \in B' \text{ such that } (a, b) \in R'.$$

This says that every animal receives exactly one classification. The words define the objects, while the notation makes the logical structure visible.

Making a Statement Sufficiently Formal. Before presenting a mathematical claim, identify the objects being discussed, define the variables and the values they may take, state any necessary assumptions, use the required quantifiers and logical connectives, and make clear exactly what is being asserted. The appropriate degree of formality depends on the purpose and audience. A classroom explanation may use more ordinary language than a formal proof, but neither should rely on avoidable ambiguity.

The notation introduced in this chapter lets us specify which objects are under discussion, how they are related, and what is being claimed about them. Its purpose is not to replace ordinary language, but to remove ambiguity when ordinary language is insufficient.

2.6 Exercises

The questions below ask you to work with the ideas developed in the chapter.

1. Classify each item as an expression, an open sentence, or a closed sentence:

- a. $3x + 5$
- b. $3x + 5 = 20$
- c. $3(5) + 5 = 20$
- d. $\forall x \in \mathbb{R}, x^2 \geq 0$

2. Translate each item as directed:

- a. The population P is greater than 500.
- b. The probability p is between 0 and 1, inclusive.
- c. Species s belongs to the set E of endangered species.

- d. Write $A \cap B = \emptyset$ in words.
- e. Write $\forall x \in D, f(f(x)) = x$ in words.
- f. The average population growth rate r is the change from N_1 to N_2 divided by the elapsed time from t_1 to t_2 .
3. Evaluate the following expression and explain which convention you use at each step:

$$20 - 12 \div 3 \times 2.$$

4. Let

$$U = \{\text{bat, eagle, penguin, whale}\},$$

and let M contain the mammals in U , while F contains the animals in U that can fly. Draw a Venn diagram for M and F , then state $M \cap F$ and $M \cup F$.

5. Let

$$A = \{\text{cat, dog, eagle}\}, \quad B = \{\text{mammal, bird}\}.$$

Write a relation from A to B that records the biological class of each animal. Is this relation also a function? Explain.

6. Complete the following truth table. Then let P mean “the organism is a dog” and Q mean “the organism is a mammal.” Write $P \Rightarrow Q$ and its converse in words.

P	Q	$\neg Q$	$P \wedge \neg Q$
T	T		
T	F		
F	T		
F	F		

7. Write the negation of each statement without simply placing the words “it is not true that” at the beginning:
- Every organism in the sample exhibits the trait.
 - There exists a patient whose temperature satisfies $T \geq 38^\circ\text{C}$.

Additional Exercises

1. Decide whether each of the following is a binary operation on the stated set:

- a. addition on $\mathbb{Z}$;
- b. subtraction on the positive integers;
- c. multiplication on $\mathbb{R}$; and
- d. division on $\mathbb{R}$.

2. Let $a \star b = a + 2b$. Calculate

$$5 \star (-2), \quad (1 \star 3) \star 2, \quad 1 \star (3 \star 2).$$

Is $\star$ associative?

- 3. Give a biological example of a set, a relation, and a function.
- 4. The sentence “Each specimen has one measurement” is ambiguous. List three pieces of information needed to turn it into a precise mathematical claim.

Closing Reflection

Choose one mathematical symbol that once seemed unfamiliar to you. Explain its meaning in an ordinary English sentence and give an example of its use.

Remaining Chapters Under Construction

This edition currently contains Chapters 1 and 2. Further chapters are being prepared and will be added to later versions of these lecture notes.

Appendix A

Solutions to Exercises

The solutions in this appendix are model answers. Other well-justified answers are possible.

A.1 Chapter 1: Mathematics in Our World

1. A useful model might include distance, transport, traffic, weather, waiting time, and road closures. The vehicle's colour could be omitted because it does not materially affect journey time.
2. The next three terms are 34, 55, and 89. The corresponding ratios are

$$\frac{34}{21} \approx 1.6190, \quad \frac{55}{34} \approx 1.6176, \quad \frac{89}{55} \approx 1.6182.$$

These support the conjecture that the ratios approach φ , but finitely many examples cannot prove a statement about the entire sequence.

3. One possible measurement is the number or area of visible yellow lesions. It captures one sign of diabetic retinopathy but may overlook vessel damage, image quality, lesion type, or location.
4. For a claim that a shell follows the golden ratio, we would need defined measurements, a reliable procedure, a suitable sample, and a stated tolerance for closeness to φ . Visual resemblance alone is not sufficient evidence.
5. Mathematics extends beyond calculation. A navigation application represents roads as a network and uses several mathematical tools to select a route; the Fibonacci and golden-ratio examples use numbers, diagrams, and logical relationships to investigate natural patterns. These illustrate practical, intellectual, and aesthetic dimensions of mathematics.
6. The developers should disclose the test data, error rates, and loss of performance on blurry images. They should state that the system supports screening rather than diagnosis and that

uncertain cases require clinical review. Integrity requires reporting assumptions, uncertainty, and known limitations—not only a favourable overall accuracy.

A.2 Chapter 2: The Language of Mathematics

1. (a) $3x + 5$ is an **expression**.
 (b) $3x + 5 = 20$ is an **open sentence** because its truth depends on the value of x .
 (c) $3(5) + 5 = 20$ is a **closed sentence**, and it is true.
 (d) $\forall x \in \mathbb{R}, x^2 \geq 0$ is a **closed sentence**, and it is true.
2. (a) $P > 500$.
 (b) $0 \leq p \leq 1$.
 (c) $s \in E$.
 (d) The sets A and B have no elements in common.
 (e) For every x in D , applying f twice returns x .
 (f)

$$r = \frac{N_2 - N_1}{t_2 - t_1}.$$

3. Division and multiplication have equal priority and are performed from left to right:

$$20 - 12 \div 3 \times 2 = 20 - 4 \times 2 = 20 - 8 = 12.$$

4. Here

$$M = \{\text{bat, whale}\}, \quad F = \{\text{bat, eagle}\}.$$

The bat belongs in the overlap, the whale in the M -only region, the eagle in the F -only region, and the penguin outside both circles. Therefore,

$$M \cap F = \{\text{bat}\}, \quad M \cup F = \{\text{bat, eagle, whale}\}.$$

5. One suitable relation is

$$R = \{(\text{cat, mammal}), (\text{dog, mammal}), (\text{eagle, bird})\}.$$

It is also a function because each element of A is paired with exactly one element of B .

	P	Q	$\neg Q$	$P \wedge \neg Q$
6.	T	T	F	F
	T	F	T	T
	F	T	F	F
	F	F	T	F

The implication $P \Rightarrow Q$ says:

If the organism is a dog, then it is a mammal.

Its converse says:

If the organism is a mammal, then it is a dog.

The implication is true, but its converse is false.

7. (a) At least one organism in the sample does not exhibit the trait.
 (b) Every patient has a temperature below 38°C .

Additional Exercises

1. Addition on $\mathbb{Z}$ and multiplication on $\mathbb{R}$ are binary operations.

Subtraction on the positive integers is not a binary operation because, for example, $2 - 3 = -1$, which is not positive. Division on $\mathbb{R}$ is not a binary operation because division by zero is undefined.

- 2.

$$5 \star (-2) = 1, \quad (1 \star 3) \star 2 = 11, \quad 1 \star (3 \star 2) = 15.$$

Because the final two results are different, $\star$ is not associative.

3. For example, let a set consist of the specimens included in a study. Let a relation record which traits each specimen exhibits, and let a function assign each specimen its recorded mass.
4. We must identify the specimens under discussion, the quantity being measured and its unit, and what “one measurement” means. For example, does it mean one reading taken at a specified time or one recorded variable?

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